

View Meta-Reviews

Paper ID

338

Paper Title

Achieving Optimal Static and Dynamic Regret Simultaneously in Bandits with Deterministic Losses

META-REVIEWER #1

META-REVIEW QUESTIONS

2. Meta-review

The reviewers agreed that the submission addresses a meaningful problem with results that are both technically and conceptually non-trivial. While one reviewer raised their score after the rebuttal clarified that an alternative Exp4-based approach was inapplicable, the overall consensus remained cautious. The discussion highlighted that the current results depend fundamentally on the assumptions of deterministic losses and a known S . Although this work represents a potential 'first step,' the reviewers concluded that it remains unclear whether the proposed techniques or algorithmic ideas can be extended to more general settings, which limited the perceived significance of the contribution.

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Paper Title

Achieving Optimal Static and Dynamic Regret Simultaneously in Bandits with Deterministic Losses

Reviewer #1

Questions

1. Rating (scale of 1-7)

4: Marginally above acceptance threshold

2. Confidence (scale of 1-4)

3: The reviewer is confident in their evaluation

3. Review

The paper taps into the investigation of bandit algorithms that perform well against multiple benchmarks of varying complexity. In particular, the regret against a single arm (static regret), and against a benchmark that may switch arms S times (dynamic regret) is investigated. The paper imposes the restriction that arm losses are constant between S switches.

Using Blackwell approachability, an algorithm is devised that simultaneously achieves near optimal static and dynamic regret against an oblivious adversary. The crucial observation for the algorithm is that negative static regret can compensate for the additional exploration needed for small dynamic regret. This is a nice idea. It remains open if the approach of the paper can be used in a more general setting.

It is known that small static and small dynamic regret cannot be achieved against an adaptive adversary, and the paper gives a nice and simple example in the restricted context of the paper.

On the downside, switching against an oblivious adversary can be handled in much more generality (unknown S , arbitrary losses) by EXP4 (Auer et al., 2002) with non-uniform weights. In the vanilla version, this requires an exponential number of weights, but these can be aggregated into $A \cdot S$ weights. The current paper needs only A weights and achieves high probability regret bounds, whereas EXP4 bounds the expected regret. I believe it is unknown if EXP4 can achieve high probability regret bounds in this setting.

I have not checked the proofs, but the results seem plausible.

Added after rebuttal:

The authors are correct that simple EXP4 does not solve the issue. I am raising my score. Still the setting is quite restrictive, begging the questions (1) if the current approach extends to more interesting settings, and (2) if a simpler (and maybe more extendible) algorithm might suffice.

Reviewer #2

Questions

1. Rating (scale of 1-7)

4: Marginally above acceptance threshold

2. Confidence (scale of 1-4)

3: The reviewer is confident in their evaluation

3. Review

This paper studies simultaneous optimality for static and dynamic regret in adversarial multi-armed bandits. The paper claims two main contributions: (i) an extension of the Marinov-Zimmert impossibility result to deterministic losses, showing that against an adaptive adversary one still cannot simultaneously obtain optimal static and dynamic regret; and (ii) a positive result against an oblivious adversary with deterministic losses and known number of switches S , where the proposed algorithm achieves $\sim O(\sqrt{AT})$ static regret and $\sim O(\sqrt{SAT})$ dynamic regret simultaneously. The technical core is an interesting Blackwell-approachability / no-regret reduction that uses negative static regret to offset the exploration needed for dynamic regret.

I found the paper technically interesting and conceptually nontrivial. The adaptive/oblivious separation is genuine, and the positive result is not vacuous: the paper also proves that even under deterministic losses and known S , the adaptive-adversary impossibility persists. The warm-up section helps in conveying the core mechanism, namely that extra exploration can be "paid for" by negative static regret when rarely played arms become good, and that approachability can hedge between the relevant regimes. The overall theorem is, to my knowledge, the first optimal simultaneous guarantee in this setting, under the paper's assumptions.

That said, I view the paper as somewhat borderline for COLT because the positive result relies on two quite restrictive assumptions at once: deterministic losses and prior knowledge of S . The paper itself explicitly notes that deterministic losses are crucial for

deriving the observable static-regret upper bound in the “dynamic” epochs, and that relaxing either assumption is left open. This makes the result feel more like an important first step than a definitive resolution of the simultaneous-optimality question. In particular, the connection to the broader “best-of-all-worlds” agenda is more suggestive than fully realized.

A second weakness is exposition. The warm-up is good, but the general algorithm is fairly hard to parse. Several clipping operations and feasibility constraints appear somewhat ad hoc on first reading, and the reader only later learns that they are needed for high-probability control. I would encourage the authors to add a short roadmap of the proof architecture and a clearer explanation of why each ingredient in DYNASORE/EXPEST is indispensable.

Overall, I am positive on the technical novelty and on the conceptual separation result, but I am not fully convinced about impact at the COLT bar because of the narrow scope of the positive theorem.

Suggestions for improvement:

1. Clarify more sharply which barrier forces deterministic losses, and whether any noisy/stochastic variant seems within reach.
2. Discuss more explicitly whether the assumption that S is known can be removed, even at the price of extra logarithmic factors.
3. Improve the presentation of the main algorithm and explain the role of the clipping/constraint choices earlier.

Reviewer #3

Questions

1. Rating (scale of 1-7)

4: Marginally above acceptance threshold

2. Confidence (scale of 1-4)

3: The reviewer is confident in their evaluation

3. Review

SUMMARY

This paper investigates whether a multi-armed bandit algorithm can simultaneously achieve optimal static regret ($\tilde{\mathcal{O}}(\sqrt{AT})$) and dynamic regret ($\tilde{\mathcal{O}}(\sqrt{SAT})$). It establishes an impossibility result for adaptive adversaries with deterministic losses (Theorem 2). However, it presents DYNASORE, an algorithm that achieves this simultaneous optimality against oblivious adversaries with

deterministic losses and a known number of switches S (Theorem 4). The algorithm successfully reduces the goal to a Blackwell approachability problem.

STRENGTHS

The formal separation between the power of adaptive and oblivious adversaries regarding simultaneously optimal regret bounds (Theorem 2 versus Theorem 4) resolves a substantial structural question in the best-of-both-worlds literature.

The algorithmic architecture is distinctive. Lifting the problem to a multi-objective approachability game and explicitly mathematically banking negative static regret (accrued from unplayed arms during aggressive exploration epochs) to finance the $\tilde{O}(\sqrt{SAT})$ dynamic regret detection cost is a structural insight that is not immediate from the problem formulation.

The use of a hybrid regularizer (combining Shannon entropy for static regret coordinates and a log-barrier for the dynamic regret coordinate) in the Optimistic Mirror Descent (OMD) meta-algorithm (Lemma 13) is a sophisticated technique precisely tailored to the asymmetric bounds of the subroutines.

WEAKNESSES

The requirement for deterministic losses and a known number of switches, S , may limit the practical applicability and overall impact of this framework.

QUESTIONS FOR THE AUTHORS

Regarding Theorem 2, the current analysis assumes the adversary can observe the learner's expected number of draws to arm 2. However, in standard settings, the adversary should only be able to observe the learner's realized number of draws to a given arm. If the adversary's observation is restricted to realized draws, does the impossibility result of Theorem 2 still hold?

If the objective is to achieve static and dynamic regret where the non-stationary measure is defined by the temporal variation of the loss sequence (rather than the number of switches), can the current analysis be generalized in a straightforward manner?

In the pseudocode of Algorithm 2 (line 11), the feasible set is defined as $\Theta = \{\theta \in \Delta_{A+1} : 1 - \theta(A+1) \geq \gamma\}$. However, on page 12, the condition $\theta(A+1) \geq \gamma$ is used. Could the authors clarify if this is a typographical error?